The Sixth Nonlinear-Eigenvalue Trace Coefficient

Radial quadratic cases in dimensions 9 and 11

Proof packet by `agent_lane_06`

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Status. Candidate partial result, likely valid, subject to human review. This settles the first uncomputed coefficient cases singled out in the source for one positive elliptic polynomial, not for all such polynomials.

1 Source problem

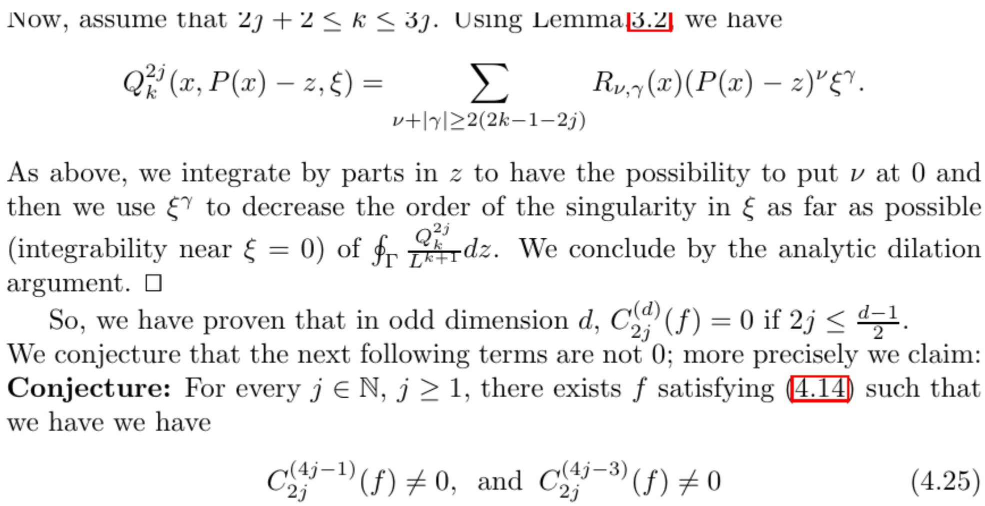
For a positive elliptic polynomial P on $\mathbb{R}^d$, Aboud and Robert study the quadratic operator pencil

$$L(\lambda) = -\Delta + (P(x) - \lambda)^2$$

and its nonlinear spectrum. Their semiclassical trace formula has coefficients $C_{2j}^{(d)}(f)$. In odd dimensions they prove $C_{2j}^{(d)}(f) = 0$ when $d \geq 4j + 1$, and conjecture that the next two coefficients do not vanish: for each $j \geq 1$, an admissible f should exist with

$$C_{2j}^{(4j-1)}(f) \neq 0, \qquad C_{2j}^{(4j-3)}(f) \neq 0.$$

The source verifies $j = 1$, treats several $j = 2$ cases, and states explicitly that $d = 9, 11$ require symbolic or numerical help to compute $C_6^{(d)}$ [1].


Now, assume that $2j+2 \leq k \leq 3j$. Using Lemma 3.2, we have

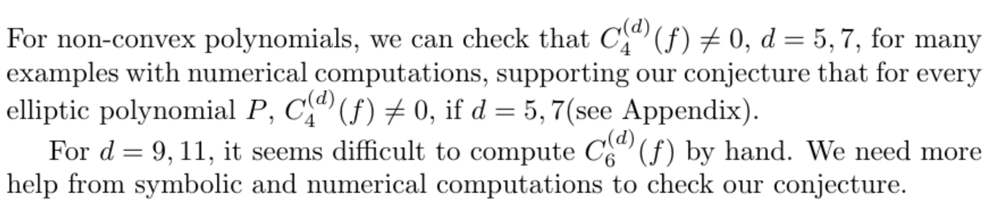
$$Q_k^{2j}(x, P(x)-z, \xi) = \sum_{\nu+|\gamma|\geq 2(2k-1-2j)} R_{\nu,\gamma}(x)(P(x)-z)^{\nu}\xi^{\gamma}.$$

As above, we integrate by parts in z to have the possibility to put ν at 0 and then we use ξ^{γ} to decrease the order of the singularity in ξ as far as possible (integrability near $\xi = 0$) of $\oint_{\Gamma} \frac{Q_k^{2j}}{L^{k+1}} dz$. We conclude by the analytic dilation argument. $\square$

So, we have proven that in odd dimension d, $C_{2j}^{(d)}(f) = 0$ if $2j \leq \frac{d-1}{2}$.
We conjecture that the next following terms are not 0; more precisely we claim:
Conjecture: For every $j \in \mathbb{N}$, $j \geq 1$, there exists f satisfying (4.14) such that we have we have

$$C_{2j}^{(4j-1)}(f) \neq 0, \text{ and } C_{2j}^{(4j-3)}(f) \neq 0 \tag{4.25}$$

Figure 1: Conjecture (4.25) in the source, PDF page 8.


For non-convex polynomials, we can check that $C_4^{(d)}(f) \neq 0$, $d = 5, 7$, for many examples with numerical computations, supporting our conjecture that for every elliptic polynomial P, $C_4^{(d)}(f) \neq 0$, if $d = 5, 7$(see Appendix).

For $d = 9, 11$, it seems difficult to compute $C_6^{(d)}(f)$ by hand. We need more help from symbolic and numerical computations to check our conjecture.

Figure 2: The source's explicit $d = 9, 11$ symbolic-computation prompt, PDF page 15.

2 Result

Theorem 1. *Let $P(x)=|x|^2$. With the source's normalization of the semiclassical trace coefficients,*

$$C_6^{(9)}((z+1)^{-15}) = \frac{212857\pi}{901943132160} > 0,$$
$$C_6^{(11)}((z+1)^{-18}) = \frac{84227\pi}{2727476031651840} > 0.$$

Consequently the pencil

$$-\Delta + (|x|^2-\lambda)^2$$

has infinitely many nonlinear eigenvalues in dimensions 9 *and* 11*. Writing $N(R)$ for the number in $|\lambda|\le R$, counted with multiplicity, for every $\varepsilon>0$ there are positive constants such that, for all sufficiently large R,*

$$N_{d=9}(R) \ge c_\varepsilon R^{9/2-\varepsilon}, \qquad N_{d=11}(R) \ge c'_\varepsilon R^{15/2-\varepsilon}.$$

3 Proof intuition

For the radial quadratic polynomial, every symbol in the Weyl-parametrix recurrence is invariant under simultaneous rotations of x and ξ. Therefore it depends only on four scalars: $a=|x|^2$, $b=|\xi|^2$, $c=x\cdot\xi$, and $w=a-z$. The apparently high-dimensional recurrence for K_6 then becomes ordinary symbolic differentiation in these four variables. Each resulting monomial can be integrated exactly: spherical averaging removes powers of c, the ξ radius gives a beta integral, the z contour gives a derivative of f, and the x radius gives a second beta integral. The final rational multiples of π are positive.

4 Invariant recurrence

Put

$$a=|x|^2, \quad b=|\xi|^2, \quad c=x\cdot\xi, \quad w=a-z, \quad \ell=b+w^2.$$

For an invariant function $F(a,b,c,w)$, differentiation with respect to a at fixed z is

$$D_aF = (\partial_a+\partial_w)F.$$

Direct differentiation gives

$$\Delta_x F = 2dD_aF + 4aD_a^2F + 4cD_a\partial_cF + b\partial_c^2F,$$
$$\Delta_\xi F = 2d\partial_bF + 4b\partial_b^2F + 4c\partial_b\partial_cF + a\partial_c^2F.$$

Define

$$S_2F = \frac{1}{4}\Delta_xF + \frac{w}{2}\Delta_\xi F + 2aF_b + 4c^2F_{bb} + 4acF_{bc} + a^2F_{cc},$$
$$S_4F = \frac{1}{16}\Delta_\xi^2F.$$

These are exactly the order-two and order-four contractions in the source's Weyl recurrence. Indeed,

$$D_x^4\ell = 8(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}),$$

which yields the displayed S_4; all mixed x, ξ derivatives of ℓ vanish. Since derivatives of ℓ of total order above four vanish, the recurrence through order six is

$$\begin{aligned}
K_0 &= \ell^{-1}, \\
K_2 &= -K_0 S_2 K_0, \\
K_4 &= -K_0(S_2 K_2 + S_4 K_0), \\
K_6 &= -K_0(S_2 K_4 + S_4 K_2).
\end{aligned}$$

As a convention check, the first identity produced by this recurrence is

$$K_2 = \frac{2dw + 4a}{\ell^3} + \frac{-4wb - 8c^2 - 8aw^2}{\ell^4},$$

which is precisely the source's displayed K_2 formula specialized to $P(x) = |x|^2$.

5 Exact integration

After putting K_6 over a common denominator, the exact recurrence gives

$$K_6 = \frac{N_d(a, b, c, w)}{2\ell^{10}},$$

where N_d has 33 monomials for each of $d = 9, 11$ and only even powers of c. For a monomial $a^A b^B c^{2k} w^W / \ell^{10}$, spherical averaging and the radial beta integral give

$$\int_{\mathbb{R}^d} \frac{b^B c^{2k}}{(b + w^2)^{10}}\, \mathrm{d}\xi = a^k \frac{(1/2)_k}{(d/2)_k} \frac{\pi^{d/2}\Gamma(B + k + d/2)\Gamma(10 - B - k - d/2)}{\Gamma(d/2)\Gamma(10)} (w^2)^{B+k+d/2-10}. \tag{1}$$

For every monomial occurring in dimensions 9 and 11, the second gamma argument in (1) is strictly positive. Thus every use of (1) is an ordinary convergent integral, not a termwise analytic regularization.

The source uses the normalized contour integral $\oint = (2\pi i)^{-1} \int$. Hence, for $r \geq 1$,

$$\oint (a - z)^{-r} f(z)\, \mathrm{d}z = \frac{(-1)^r}{(r-1)!} f^{(r-1)}(a).$$

Finally,

$$\int_{\mathbb{R}^d} a^p (a + 1)^{-q}\, \mathrm{d}x = \frac{\pi^{d/2}\Gamma(p + d/2)\Gamma(q - p - d/2)}{\Gamma(d/2)\Gamma(q)}.$$

Applying these formulas term by term, including the trace prefactor $-2(2\pi)^{-d}$, gives the following grouped contributions. Here the row (p, r) collects all terms proportional before radial integration to $a^p f^{(r)}(a)$.

d	(p,r)	contribution
9	$(0,0)$	$-11951\pi/33822867456$
9	$(1,1)$	$1802663\pi/789200240640$
9	$(2,2)$	$-6605027\pi/3156800962560$
9	$(3,3)$	$7528807\pi/18940805775360$
11	$(2,0)$	$84227\pi/454579338608640$
11	$(3,1)$	$-84227\pi/545495206330368$

Summing the $d = 9$ rows and the $d = 11$ rows gives the two constants in Theorem 1.

The chosen functions are admissible: because P has degree $m = 2$, the source requires $\mu > 3d/2$, and $15 > 27/2$, $18 > 33/2$. The source's trace criterion therefore implies that the nonlinear spectrum is nonempty; its counting proposition and scaling corollary in fact give infinitely many eigenvalues and the exponents

$$(d-2j)\frac{m+1}{m} = (d-6)\frac{3}{2},$$

which are $9/2$ and $15/2$. This proves the theorem. □

6 Verification and reproducibility

The script `code/verify_c6_radial.py` performs the recurrence and all integrals in exact SymPy arithmetic. It asserts the specialized source formula for K_2, checks the convergence condition for every monomial, and asserts both final constants exactly. The command

```
conda run --no-capture-output -n sandbox python code/verify_c6_radial.py
```

ends with `All exact assertions passed.` The complete recorded output is in `verification.md`.

7 Limitations and novelty check

This proves Conjecture (4.25) only for $j = 3$ and the radial quadratic polynomial in its two critical odd dimensions. It does not handle arbitrary positive elliptic P, dimensions arising from $j \geq 4$, or completeness of the generalized eigenfunctions.

A bounded novelty search on 22 July 2026 checked the run's four cheap indexes and exact web/arXiv searches using the source title, the coefficient notation $C_{2j}^{(4j-1)}$, the phrases "positive elliptic polynomial" and "nonlinear eigenvalue", and the dimensions $9, 11$. It found the source, the earlier Helffer–Robert–Wang work, and later numerical work on quadratic pencils, but no analytic computation of these two coefficients or theorem implying the displayed result. Novelty confidence is therefore moderate, not definitive, because this is a niche literature and notation-based search can miss differently phrased results.

8 Human-review recommendation

Check first the Weyl-product normalization in S_2, S_4 and the sign of the normalized contour residue. The exact match of K_2 with the source is the main convention audit. Then rerun the verifier and inspect the two grouped sums. If those checks pass, the result is ready to retain as a rigorous partial solution.

References

[1] F. Aboud and D. Robert, *Asymptotic expansion for nonlinear eigenvalue problems*, J. Math. Pures Appl. 93 (2010), 149–162, arXiv:0903.0919, https://arxiv.org/abs/0903.0919.